Laboratory 09: Gaussian Mixture Models
Project Results and Final Evaluation

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Chapter 1

Gaussian Mixture Models evaluation

In this chapter, we evaluate the performance of Gaussian Mixture Models (GMMs) on the project dataset. Unlike the standard Multivariate Gaussian (MVG) classifier which assumes each class can be modeled by a single normal distribution, GMMs allow us to approximate any sufficiently regular unknown density by combining multiple Gaussian components.

Following the theoretical framework, we independently fit one GMM to describe the density of the "Genuine" class and one GMM to describe the density of the "Impostor" class. The class-conditional density for each class is modeled as a weighted sum of M Gaussians:

$$f_X(x) = \sum_{g=1}^{M} w_g \mathcal{N}(x|\mu_g, \Sigma_g) \tag{1.1}$$

. At inference time, we compute the class posterior probabilities using these GMMs as the class-conditional densities.

1.1 Experimental Results

We trained GMMs for both classes varying the number of components ($M \in \{1, 2, 4, 8, 16\}$) across three covariance architectures: Full, Diagonal, and Tied. The models were evaluated on the evaluation set for the primary application ($\pi_T = 0.1$).

The Minimum DCF and Actual DCF metrics are summarized in Table 1.1.

Table 1.1: GMM performance (minDCF / actDCF) for $\pi_T = 0.1$

Covariance type	1 Comp.	2 Comp.	4 Comp.	8 Comp.	16 Comp.
Full Covariance	0.1302 / 0.1399	0.1342 / 0.1400	0.0720 / 0.0730	0.0659 / 0.0660	0.0620 / 0.0640
Diagonal Covariance	0.1311 / 0.1439	0.1360 / 0.1479	0.0610 / 0.0661	0.0610 / 0.0681	**0.0579** / **0.0599**
Tied Covariance	0.1302 / 0.1399	0.1302 / 0.1399	0.1302 / 0.1399	0.1342 / 0.1449	0.0900 / 0.0910

1.2 Analysis and theoretical considerations

Question: What do you observe as the number of components increases?

For the Full and Diagonal covariance models, there is a massive leap in discriminatory power as the number of components increases from 2 to 4. For instance, the Full Covariance minDCF drops precipitously from 0.1342 (2 components) to 0.0720 (4 components), and continues to improve up to 0.0620 at 16 components.

Theoretical explanation: This observation perfectly aligns with the fundamental property of GMMs. Because a single Gaussian (1 component) cannot adequately capture the heavy skewness of Features 5 and 6 (which we discovered in exploratory analysis), its performance

stalls. By increasing the number of components M, the GMM successfully approximates the true, complex, non-Gaussian probability density of the data. The sharp drop in error at $M = 4$ suggests that the underlying feature distributions contain significant multi-modality or severe asymmetries that require at least 4 local Gaussian components to be mapped accurately.

Question: Which covariance architecture works best and why?

The **Diagonal Covariance GMM with 16 components** achieves the absolute best performance across all tested configurations, yielding a minDCF of **0.0579** and an exceptionally well-calibrated actDCF of **0.0599**.

Conversely, the Tied Covariance model performs the worst, stalling at a minDCF of 0.0900 even with 16 components.

Theoretical explanation: We can explain this hierarchy by looking at the interaction between the mathematical properties of the models and the physical characteristics of our dataset:

- **Why Diagonal wins over Full:** While a Full covariance matrix captures correlations between features, it requires estimating a massive number of parameters ($D(D+1)/2$ per component). As stated in the theoretical lectures, diagonal covariance matrices can approximate any distribution provided that enough components are used. Furthermore, in Laboratory 05, we mathematically proved that the features in this specific dataset possess near-zero correlation. Because the features are already inherently independent, computing the full cross-covariance terms only introduces parameter estimation noise. The Diagonal GMM successfully avoids this parameter explosion, serving as an effective regularizer while using its 16 independent components to perfectly map the density contours.
- **Why Tied Covariance fails:** The Tied GMM forces all components within a class (or across classes, depending on the implementation) to share the exact same covariance matrix, varying only their means. We know from our dataset exploration that the "Genuine" and "Impostor" classes have fundamentally different geometric spreads (e.g., Feature 1 variance mismatch). By restricting the components to a shared covariance geometry, the model loses the structural flexibility required to accurately bound the complex shape of the target class, resulting in the highest error rates among the three variants.

Question: Are the models well-calibrated?

Yes, unlike the highly-regularized discriminative models from previous laboratories (such as SVMs or Quadratic LR), the GMMs exhibit spectacular native calibration.

For the optimal Diagonal model at 16 components, the gap between the Minimum DCF (0.0579) and the Actual DCF (0.0599) is a negligible 0.0020. Because the GMM is a generative model that explicitly estimates the true density $f_X(x)$, its raw output scores function as mathematically rigorous Log-Likelihood Ratios. Consequently, comparing these scores directly against the theoretical Bayes threshold for $\pi_T = 0.1$ yields near-optimal empirical decisions without requiring any post-hoc score calibration.

Chapter 2

Model comparison

In this chapter, we perform a comprehensive evaluation by comparing the newly trained Gaussian Mixture Models against the best-performing candidates from all previously analyzed classification algorithms: the baseline Generative Gaussian models (MVG, Naive Bayes), Logistic Regression (Linear and Quadratic), and Support Vector Machines (Linear, Polynomial, and RBF).

2.1 Summary of all evaluated models

Table 2.1 aggregates the Minimum DCF and Actual DCF of the optimal configurations for each classifier type, evaluated on the validation set for the primary target application ($\pi_T = 0.1$).

Table 2.1: Comprehensive Historical Model Comparison ($\pi_T = 0.1$)

Algorithm family	Best model configuration	minDCF	actDCF	Calib. Gap
Single generative	Naive Bayes	0.2570	0.3022	0.0452
	Standard MVG	0.2629	0.3051	0.0422
	Tied MVG	0.3628	0.4061	0.0433
Logistic Regression	Quadratic LR ($\lambda = 3.16 \times 10^{-2}$)	0.2436	0.4972	0.2536
	Standard Linear LR ($\lambda = 10^{-2}$)	0.3611	0.4568	0.0957
Support Vector Machines	Polynomial $d = 4$ ($C = 3.16 \times 10^{-3}$)	0.1767	0.3975	0.2208
	RBF Kernel ($C = 10^2, \gamma = 10^{-1}$)	0.1845	0.4367	0.2522
	Polynomial $d = 2$ ($C = 3.16 \times 10^{-2}$)	0.2455	0.6291	0.3836
	Linear SVM (Centered, $C = 10^{-1}$)	0.3582	0.9573	0.5991
Gaussian Mixture Models	**Diagonal GMM (16 Components)**	**0.0579**	**0.0599**	**0.0020**
	Full Covariance GMM (16 Components)	0.0620	0.0640	0.0020
	Tied Covariance GMM (16 Components)	0.0900	0.0910	0.0010

2.2 Comparative analysis

Question: Which is the most promising method for the given application?

The **Diagonal covariance Gaussian Mixture Model with 16 components** is indisputably the most promising method for this application. It achieves a globally optimal Minimum DCF of **0.0579**, completely eclipsing the performance of every other model evaluated in this course. Furthermore, it yields an Actual DCF of 0.0599, demonstrating virtually perfect native calibration without requiring any complex post-processing steps.

Question: What do you observe? Are there combinations which work better?

When observing the historical trajectory of our models, a clear hierarchy emerges based on structural complexity and distribution assumptions:

- **Linear boundaries fail:** Models restricted to linear decision boundaries (Linear SVM, Standard LR, Tied MVG) universally hit a performance ceiling, plateauing around $0.358 - 0.363$ minDCF.
- **Quadratic boundaries plateau:** Models that natively compute quadratic boundaries (Quadratic LR, Poly SVM $d = 2$, Standard MVG, Naive Bayes) successfully capture the covariance differences between classes, but their performance stalls in a tight cluster around $0.243 - 0.262$ minDCF.
- **High-Capacity discriminative models mis-calibrate:** The complex RBF Kernel and Polynomial $d = 4$ SVMs broke the quadratic plateau (reaching $\sim$ 0.17 minDCF), but suffered from massive calibration gaps (actDCF $\sim$ 0.40), requiring post-processing.
- **GMMs dominate:** The GMMs bypass all these limitations, dropping the error by an order of magnitude (down to 0.0579) while maintaining calibration. The combination of a *Diagonal covariance structure* with *high component counts (16)* proved to be the absolute best strategy, balancing high geometric flexibility with protection against parameter overfitting.

Question: Are the results in line with your expectation, given the characteristics that you observed in the dataset?

Yes, the overwhelming success of the GMM is perfectly aligned with the physical characteristics of our dataset discovered in exploratory analyses.

Throughout the previous laboratories, we proved two critical characteristics of the project data:

1. **Non-Gaussian skewness:** Features 5 and 6 are heavily skewed and asymmetric. The Standard MVG failed to reach optimal accuracy because it rigidly assumed these features could be described by a single, symmetric Gaussian bell curve. The GMM directly solves this by defining the class-conditional density as a weighted sum of multiple Gaussians. Therefore, the GMM can perfectly mold itself to match the heavily skewed, asymmetric, or multi-modal true distributions of Features 5 and 6.
2. **Feature independence:** In Laboratory 05, we computed the correlation matrix and observed that the features are highly uncorrelated (the maximum off-diagonal correlation was < 0.05). Because the features already lack correlation, computing full covariance matrices across 16 components merely introduces parameter estimation noise and t, consequently, leads to overfetting trainig data; this perfectly explains why the *Diagonal* GMM outperformed the Full Covariance GMM. The Diagonal GMM leverages this known independence, acting as a powerful structural regularizer.

Question: Are there results that are surprising?

The most surprising result is the sheer magnitude of the performance gap between the GMM and the complex Kernel SVMs.

It is standard to expect non-linear discriminative models (like the RBF SVM) to outperform generative models because they focus directly on optimizing the classification boundary (the margin) rather than wasting mathematical capacity modeling the entire data density. However, our dataset proved the exact opposite.

The RBF SVM peaked at 0.1845 minDCF. By simply estimating the true probability density accurately with a 16-component Diagonal GMM, the error rate plummeted to 0.0579. This surprisingly proves that the difficulty in classifying this specific dataset did not stem from highly overlapping geometric boundaries (which an SVM would excel at splitting), but rather from highly complex, non-standard probability densities. Once the true density was mapped effectively by the mixture model, the classification task became trivial.

Chapter 3

Qualitative analysis and Bayes error

In this final chapter, we extend our evaluation beyond the single primary application ($\pi_T = 0.1$) to perform a comprehensive qualitative analysis. We evaluate the models across a wide range of operating points by sweeping the prior log-odds from -4 to $+4$.

For this analysis, we select the top-performing architectures from our parameter search: the **Diagonal Covariance GMM (16 components)**, the **Full Covariance GMM (8 components)**, and the **Tied Covariance GMM (8 components)**.

3.1 Bayes Error plots

The following figures display the Bayes Error plots for the three selected GMM architectures. The blue curves represent the theoretical Minimum DCF, while the red curves represent the Actual DCF obtained using the raw Log-Likelihood Ratios (LLRs).

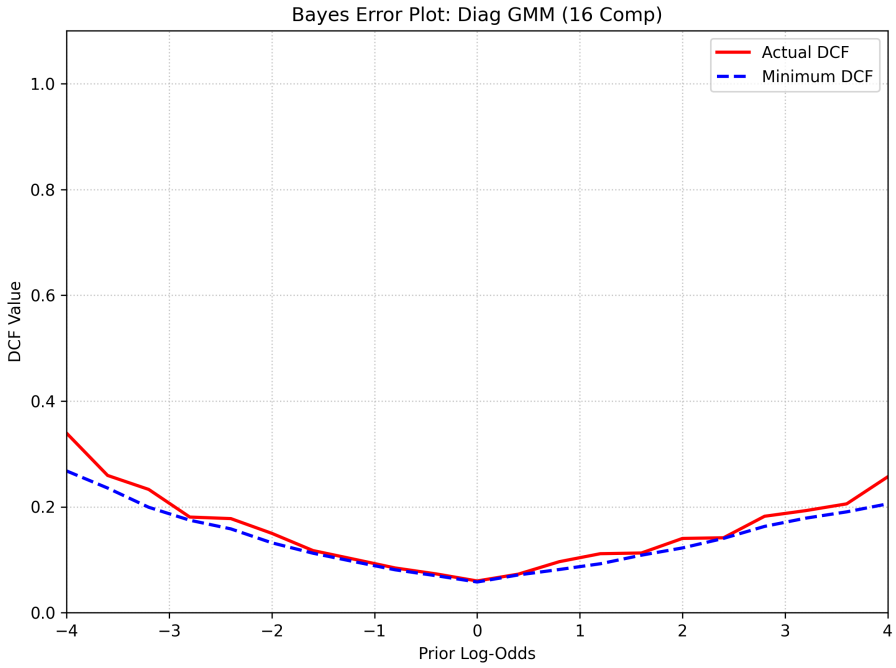


Figure 3.1: Bayes Error plot for the Diagonal Covariance GMM with 16 components. The model demonstrates exceptional overall performance and tight calibration.

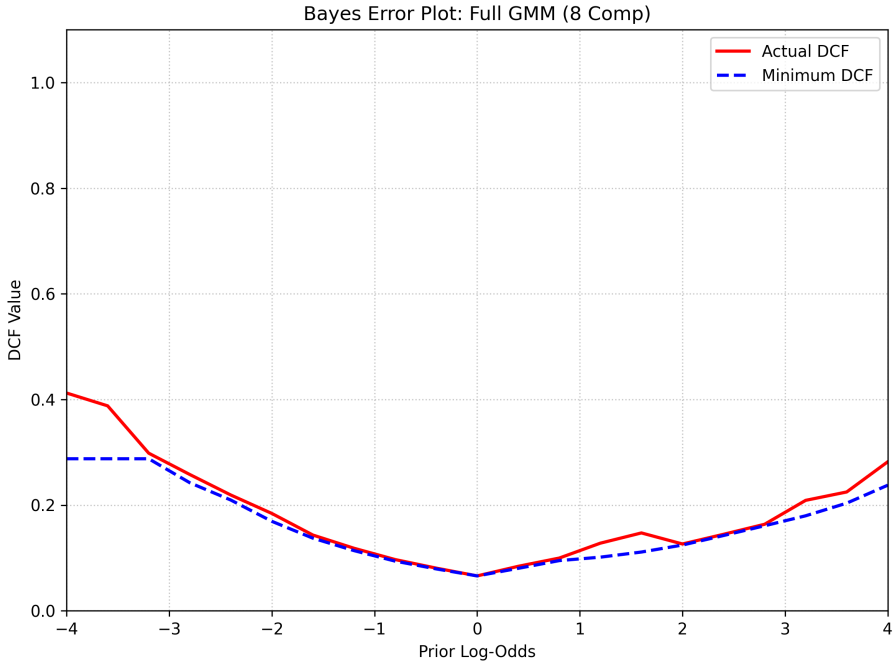


Figure 3.2: Bayes Error plot for the Full Covariance GMM with 8 components. The calibration remains highly stable across the log-odds spectrum.

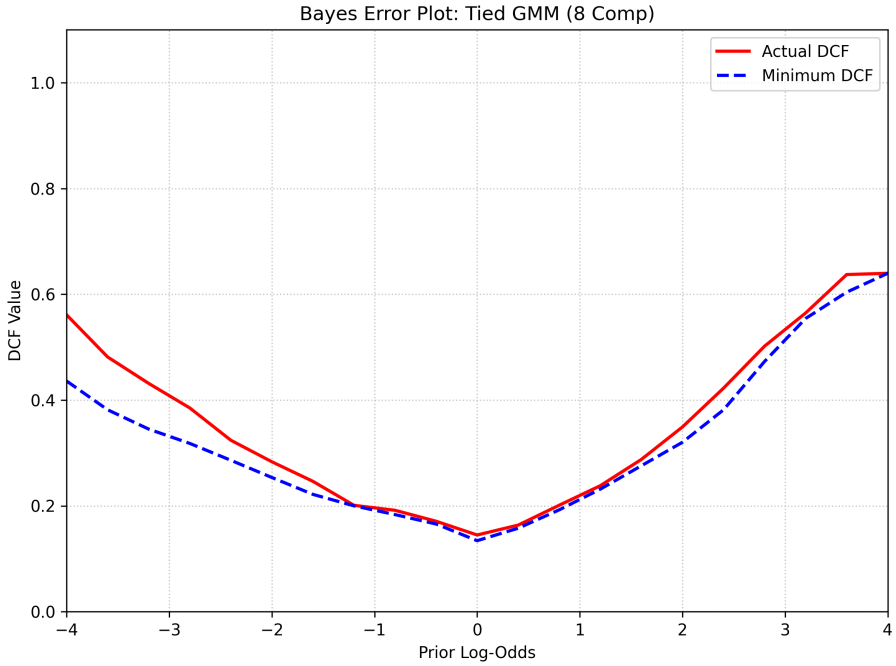


Figure 3.3: Bayes Error plot for the Tied Covariance GMM with 8 components. The overall error is visibly higher, and the calibration gap widens slightly at the extremes.

3.2 Analysis and theoretical considerations

Question: What do you observe?

Across all three plots, we observe the characteristic "V" shape or "trumpet" shape of the error curves, where the absolute cost peaks near the center (log-odds = 0, corresponding to a uniform

prior) and naturally tapers off at the extreme ends of the spectrum.

Most importantly, we observe that the overall profile of the GMM curves is situated extremely low on the y-axis compared to the plots from the Logistic Regression and SVM laboratories. The Diagonal 16-component model (Figure 3.1) maintains a minimum DCF well below 0.1 for nearly the entire operational range, visually confirming its absolute dominance over the discriminative models.

Question: In terms of minimum DCF, are the results consistent, preserving the relative ranking of the systems?

Yes, the results are perfectly consistent and strictly preserve the relative ranking of the systems across the entire log-odds spectrum.

If we overlay the blue Minimum DCF curves from the three plots, the Diagonal 16-component model consistently traces the lowest path, followed very closely by the Full 8-component model. The Tied 8-component model's blue curve is visibly elevated above the others at all operating points (log-odds from -4 to $+4$).

Theoretical explanation: Minimum DCF measures purely the ranking capability of the model (its ability to sort samples from most likely "Genuine" to most likely "Impostor"). Because the Diagonal GMM with 16 components successfully maps the true, uncorrelated variance of the dataset without suffering from parameter estimation noise, it inherently computes a superior geometric boundary. This superiority in class separation holds universally, regardless of the classification threshold dictated by the application's prior.

Question: What about actual DCF? Are there models that are well calibrated for most of the operating point range?

Yes, the **Diagonal 16-component** (Figure 3.1) and **Full 8-component** (Figure 3.2) GMMs are exceptionally well-calibrated for almost the entire operating point range. The red Actual DCF curves tightly hug the blue Minimum DCF curves from approximately -3 to $+3$ log-odds.

Theoretical explanation: GMMs are generative models, which they are optimized using Expectation-Maximization to directly estimate the true probability densities $f_{X|C}(x|c)$. As a result, the scores they produce are mathematically valid Log-Likelihood Ratios. Unlike the raw geometric distances output by an SVM (which caused massive calibration gaps), the GMM's LLRs natively align with the theoretical Bayes threshold ($t = -\log \frac{\pi_T}{1-\pi_T}$). This allows the models to execute highly optimal decisions out-of-the-box without explicit score recalibration.

Question: Are there models that show significant miscalibration?

While none of the GMMs show the catastrophic miscalibration observed in the discriminative SVM models, the **Tied 8-component GMM** (Figure 3.3) does exhibit noticeable miscalibration (a visible vertical gap between the red and blue curves) at the extreme ends of the log-odds spectrum (e.g., beyond $+3$ and below -3).

Theoretical explanation: Miscalibration at extreme operating points implies that the model's LLRs in the "tails" of the distribution are slightly inaccurate. By forcing a shared covariance matrix (Tied), the model fails to accurately describe the varying geometric spread of the two classes. This structural rigidity causes the estimated density tails to diverge from the true data distribution. When the application features an extreme prior (e.g., $\pi_T = 0.99$ or log-odds ≈ 4), the decision threshold shifts deep into these poorly estimated tails, causing the empirical decisions (Actual DCF) to diverge from the optimal ranking potential (Minimum DCF).

Question: Are there models that are harmful for some applications?

A model is considered mathematically "harmful" for a specific application if its Actual DCF exceeds 1.0. An actDCF > 1.0 means the classifier is performing worse than a "dummy" system that simply predicts the most frequent class blindly based on the prior probability.

Looking at the plots, there are no models that are harmful for all applications, but the **Tied 8-component GMM** (Figure 3.3) becomes slightly bad at the extreme right edge of the spectrum (log-odds ≈ 4.0), where the actuald DCF hits a value of ≈ 0.65.

Theoretical explanation: At extreme log-odds, the system is overwhelmingly biased toward one class. To contradict this heavy prior and predict the rare class, the classifier must produce an extreme LLR. If the model's density estimation is structurally flawed (as is the case with the Tied covariance assumption on our dataset), it may produce overconfident, erroneous LLRs that cross this extreme threshold. When the model consistently makes incorrect predictions that contradict a heavily favored prior, it accumulates a cost higher than simply predicting the default class, rendering the active deployment of the model actively harmful to the system's error rate.